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RESEARCH ARTICLE

EffNet-SVM: A Hybrid Model for Diabetic Retinopathy Classification Using Retinal Fundus Images

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ABSTRACT The manual diagnosis of diabetic retinopathy (DR) is often invasive, time-consuming, expensive, and prone to human error. Additionally, it can be subjective, depending on the clinician's professional experience. Recently, automated computer-aided diagnosis (CAD) systems have significantly reduced the time and effort required for diagnosis while achieving superior performance compared to traditional methods. Researchers have extensively explored deep learning (DL) and convolutional neural networks (CNNs) for diagnosing DR from fundus images, yielding promising results and offering a viable alternative to conventional diagnostic approaches. In this study, a hybrid model named EffNet-SVM is proposed for the classification of DR and no DR cases using retinal fundus images. The model is trained and tested using the Asia Pacific Tele-Ophthalmology Society (APTOS) dataset, which includes both DR and no DR images. The EffNet-SVM utilizes EfficientNetV2-Small for feature extraction from input fundus images, and the extracted features are then classified using a support vector machine (SVM) with a radial basis function (RBF) kernel. The EffNet-SVM model outperformed eight state-of-the-art DL models from the literature, achieving the highest accuracy of 97.26%. Performance metrics validate that the proposed hybrid model can be effectively integrated into CAD systems for the automated analysis of fundus images.

INDEX TERMS Diabetic retinopathy, deep features, EfficientNet, hybrid CNN, support vector machines, transfer learning.

I. INTRODUCTION

Diabetic retinopathy (DR) is a serious health concern affecting millions of people worldwide, especially in low-income and middle-income countries [1]. It damages retinal blood vessels due to high blood sugar levels. If not diagnosed early, DR can lead to blindness or severe vision impairment, particularly in adults. Early identification and prompt treatment are crucial to halt its progression [2].

In conventional diagnostics, an ophthalmologist analyzes fundus images, which are high-resolution photographs of the retina. These images offer a visual representation of the retinal structure, aiding the expert ophthalmologist in identifying pathological features associated with the progression of DR. The diagnosis primarily relies on recognizing characteristic lesions in the retina that indicate varying severity levels of the disease. These lesions include: (a) Microaneurysms (MAs), which appear as tiny red dots caused by the weakening of capillary walls; (b) Hemorrhages (HAs), which manifest as dark red or blot-like spots resulting from the rupture of retinal blood vessels; (c) Hard exudates (HEs), which are yellowish-white spots near the macula due to deposits of lipids and proteins leaked from damaged blood vessels; (d) Soft exudates, which arise from ischemia (lack of oxygen)

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in the nerve fiber layer and appear as fluffy white patches due to swelling; (e) Neovascularization (NV), which occurs due to the growth of abnormal new blood vessels in response to retinal ischemia; and (f) Macular edema, caused by swelling in the macula due to fluid leakage, which can lead to permanent vision loss [3].

Manual grading by ophthalmologists is a labour-intensive and time-consuming process that is susceptible to inter-observer variability, making it challenging to screen large populations effectively. Consequently, there is a critical need for automated, precise, and efficient diagnostic techniques [4]. Recently, artificial intelligence (AI), particularly deep learning (DL) and convolutional neural networks (CNNs) has emerged as a promising alternative for the efficient and accurate diagnosis of DR [5].

DL models, particularly CNNs, play a vital role in extracting visual features from fundus images to differentiate between normal and DR cases. Unlike conventional machine learning approaches that require handcrafted feature engineering, CNNs automatically learn hierarchical representations from the fundus images, capturing low-level textural features like vessel structures to high-level pathological features like hemorrhages and exudates [6]. Visual representation plays a critical role in DR classification by enabling the model to recognize fine-grained differences between varying severity levels of DR. Studies have shown that well-trained DL models can identify clinically relevant regions, for interpretability and decision-making for ophthalmologists [7].

A significant challenge in applying AI to DR identification is the requirement for large and diverse datasets. AI models, particularly DL architectures such as CNNs, require substantial amounts of labelled data for effective training [8]. The quality of these datasets directly influences the performance of the AI models. Ensuring diversity within these datasets is crucial because it enables the models to generalize across various populations, accounting for differences in age, ethnicity, and other demographic factors [9]. Large-scale initiatives, such as the Asia Pacific Tele-Ophthalmology Society (APTOS) dataset [10], provide the data necessary to develop robust AI systems for diagnosing DR from retinal fundus images. Interpretability can be enhanced through techniques such as heatmaps, which visually highlight the regions of the retinal image emphasized by the model during diagnosis [11]. These visual representations enable clinicians to identify the features that AI models consider when making the decision, thereby making the diagnostic process more transparent and trustworthy.

A. RELATED WORK

In the literature, researchers have used custom-made CNNs, advanced variants of pre-trained CNN models with transfer learning (TL), hybrid approaches, and ensemble models for the early and accurate diagnosis of DR.

Shimpi et al. [12] proposed a hybrid DR neural network (DRNN) that utilized two state-of-the-art architectures ResNet-152 [13] and DenseNet-121 [14] to enhance performance on image classification tasks. The dataset, consisting of 2,750 retinal fundus images, was categorized into five groups and assembled from Kaggle. Ensemble networks [15], [16] produced better accuracy than individual networks, and the hybrid model achieved an accuracy of 96.91%. Rao et al. [17] used ResNet-50 [13] on the EyePACS dataset [18]. Their preprocessing steps included padding the images, circular cropping to include only the circular part of the eye, and applying Gaussian blur. They achieved an accuracy of 94.15%.

Das et al. [19] conducted a comprehensive study of 26 state-of-the-art DL networks and found that EfficientNetB4 [20] achieved the highest validation accuracy of 79.11% on the Kaggle's EyePACS fundus image dataset [18]. Goel et al. [21] utilized the VGG16 model on the Indian DR Image Dataset (IDRiD) [22], which comprises 516 high-resolution images with detailed annotations of DR severity. Their proposed framework achieved an accuracy of 93.40%. Wiratama et al. [23] used a fine-tuned MobileNetV2 architecture known for its lightweight and mobile-friendly design. They utilized the APTOS 2019 dataset [10], which contains 3662 retinal fundus images divided into five classes: no-DR, mild, moderate, severe, and proliferative DR and reported a testing accuracy of 92.60%.

Mohanty et al. [24] proposed a hybrid network that combined the VGG16 [25] architecture with the XGBoost classifier [26] using the APTOS 2019 dataset [10]. Their hybrid network achieved an accuracy of 79.50%. Taifa et al. [27] introduced a novel approach to DR detection by combining machine learning algorithms with deep feature extraction techniques using DL models. They achieved a multi-class accuracy of 95.50% on the APTOS 2019 dataset [10] by stacking diverse classifiers, including Decision Trees [28], Random Forests [29], and Support Vector Machines (SVMs) [30]. Taufiqurrahman et al. [31] proposed a hybrid and efficient MobileNetV2-SVM model to classify DR, achieving a combined weighted accuracy of 85% on the APTOS 2019 dataset [10]. Mandiga et al. [32] fine-tuned the MobileNetV2-SVM model to obtain an improved weighted accuracy of 86% using the same dataset. Kobat et al. [33] utilized DenseNet for feature extraction and employed a cubic SVM to classify DR using the APTOS 2019 dataset [10], reporting an accuracy of 85.93%. Padmanayana and Anoop [34] designed a custom-made CNN for DR classification. They evaluated their CNN using the APTOS 2019 dataset [10] and a private dataset and reported accuracies of 94.60% and 94.80% on the APTOS [10] and private datasets, respectively.

Feature-based attention (FBA) has been extensively used in DL to improve classification by enhancing discriminative features while filtering out irrelevant ones. Lindsay [35] introduced a biologically inspired FBA mechanism in CNNs, demonstrating that selectively modulating category-specific feature maps improves object detection accuracy. Yu et al. [36] extended this approach by developing a parallel feature expansion model, which integrates FBA to enhance feature richness and classification robustness. More recently, Flores-Benites et al. [37] proposed TVAnet, an attention-driven model combining spatial and FBA, significantly improving visual perception in self-driving cars. Inspired by this, Liu [38] proposed an automatic grading system for DR in colour fundus images using a cascaded hybrid attention network, emphasizing explainability, and achieving 96.34% accuracy on the MESSIDOR dataset [39]. Similarly, Romero-Oraa et al. [40] introduced a novel DL approach for DR classification, leveraging an Xception-based CNN and an attention mechanism to separately focus on red and bright lesions. Their model attained 83.7% accuracy on the Kaggle EyePACS dataset [18], demonstrating enhanced feature extraction and interpretability through attention maps, though computational overhead and misclassification of certain DR severity levels remain key limitations. Attention-based architectures, while valuable for interpretability, face significant practical challenges in scaling and efficiency. Studies highlight that these models often require extensive datasets to achieve robust performance, as smaller or noisier data can lead to unreliable attention patterns [41]. In clinical settings with limited annotated data, attention mechanisms struggled to generalize effectively, underscoring their dependency on large-scale training [41]. This limits their deployment in real-time applications like medical diagnostics where latency is critical [42].

B. MOTIVATION AND NOVELTY

Literature highlights the advantages of hybrid models and the efficacy of lightweight architectures. EfficientNetV2 builds on the success of EfficientNet by optimizing both accuracy and efficiency, making it suitable for high-resolution medical images. Its scalable architecture can be adjusted to balance the trade-offs between computational cost and model accuracy, making it adaptable to various deployment scenarios, including mobile devices. EfficientNetV2's robust feature extraction capabilities can be effectively combined with SVMs, which are known for their strong classification performance, to enhance the overall model accuracy. This approach aims to improve diagnostic accuracy and provide a scalable and efficient solution for the accurate diagnosis of DR from retinal fundus images.

The novelty of our work lies in modifying EfficientNetV2-Small for feature extraction and leveraging these extracted features to train an SVM classifier. This approach achieves superior results compared to the conventional softmax function used in the CNN head for classification. To the best of our knowledge, this is the first study to employ EfficientNetV2-Small in combination with SVM for DR classification from fundus images.

C. ORGANIZATION OF THE PAPER

The rest of the sections are organized as follows: Section II provides an in-depth explanation of the EffNet-SVM architecture, detailing the training strategy employed, dataset preparation, and the hardware and software requirements for experimental simulations. Section III compares the proposed EffNet-SVM model with eight state-of-the-art DL-based models for DR classification using fundus images, as reported in the literature. The performance of EffNet-SVM is evaluated using widely recognized metrics, including the confusion matrix, accuracy, precision, recall, Receiver Operating Characteristic (ROC) curve, and Area Under the Curve (AUC). Finally, Section IV summarizes the key technical contributions and research findings of this study, along with suggestions for future research directions.

II. METHODOLOGY

This study employs three distinct models and compares their outcomes while benchmarking against the best results reported in existing literature. Image rescaling, rotation, flipping, translation, and contrast adjustments were implemented to enhance the dataset variability. These preprocessing steps were followed by training and classification. All models were built using the PyTorch framework and executed on an Nvidia Tesla V100 GPU with a maximum of 21GB RAM.

1) The first model is the fine-tuned modified EfficientNet designed from the baseline EfficientNetV2- Small, with classification performed by a softmax function.
2) The second model is an SVM classifier using a radial basis function (RBF) kernel.
3) The third model is a hybrid model that combines the modified EfficientNet with an SVM classifier featuring an RBF kernel.

Fig. 1 depicts the block diagram of various approaches proposed to classify DR from fundus photographs.

A. DATASET

The fundus images used for training and testing the models are sourced from the APTOS 2019 dataset [10]. This dataset comprises 3,662 images categorized into five groups based on the DR grading system: no DR, mild DR, moderate DR, severe DR, and proliferative DR. The images were collected over an extended period in diverse settings and under varying conditions.

For clinicians, the primary concern is often distinguishing between DR and no DR rather than classifying DR severity or grading. Additionally, Venugopal et al. [43] have demonstrated that binary classification models exhibit better generalizability and perform better than multi-class classification models. Moreover, multi-class classification models are more complex and susceptible to class imbalances. Obtaining a balanced multi-class fundus dataset with a large number of images is challenging. This motivated us to design a binary classification model rather than a multi-class one. The original five-class dataset is reorganized for binary classification to simplify the diagnostic task and focus on detecting the presence or absence of DR, grouping the data into class 0 (no DR) and class 1 (DR). The resulting dataset

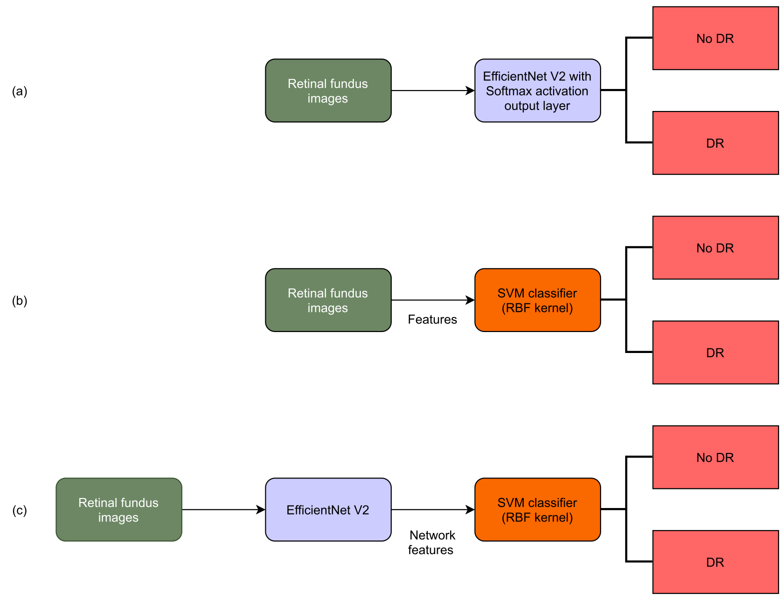

FIGURE 1. (a) Fine-tuned EfficientNetV2-Small with softmax output; (b) SVM classifier with RBF kernel; (c) Hybrid model called EffNet-SVM.

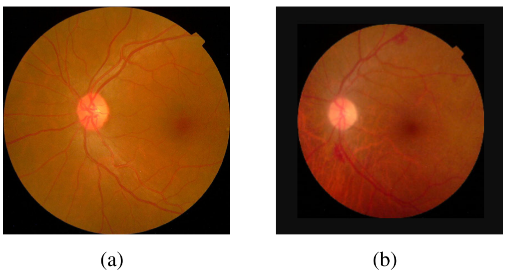

FIGURE 2. Fundus photograph of (a) no DR and (b) DR.

is relatively balanced, containing 1,805 images for no DR and 1,857 images for DR. Due to the heterogeneous origins and conditions under which the images were captured, their sizes are non-uniform. To address this, all images are rescaled to a standardized size of 224 × 224 pixels.

Fig. 2 shows two sample images from the dataset. Fig. 2(a) illustrates a fundus image without DR, whereas Fig. 2(b) depicts a fundus image with DR.

B. IMAGE AUGMENTATION

Image augmentation generates new training images through diverse transformations such as random rotations, shifts, zooms, and flips. These alterations introduce variability, enhancing the model's robustness and ability to generalize to new images [43], [44]. By expanding the dataset, image augmentation helps to reduce overfitting and improve the model's performance with diverse inputs [45]. The augmented data offers the model exposure to a broader range of variations, equipping it to perform better in real-world scenarios.

TABLE 1. "Additional layers integrated into EfficientNetV2-Small to build the modified EfficientNet model.

No.	Add-on layers
1	Global average pooling (GAP)
2	Batch normalization (BN)
3	Dropout [ratio 0.2]
4	Fully-connected (FC) + Softmax classifier

An image augmentation pipeline is employed during model training by incorporating several transformations to enhance variability. This pipeline includes random rotations of 90°, 180°, and 270°, as well as random zooming in and out. Additionally, the images were randomly flipped either horizontally or vertically, and the contrast was adjusted randomly. These operations were integrated into the training pipeline to introduce variability, thereby improving the robustness and generalization capabilities of the model. Fig. 3 illustrates the various image augmentations performed during the training phase of the proposed models.

C. MODIFIED EFFICIENTNET

In the first method, a modified EfficientNet model is designed from the baseline EfficientNetV2-Small and is trained using TL. This involves adding a global average pooling (GAP) layer, followed by batch normalization (BN) and a dropout layer with a probability of 0.2. The fully-connected (FC) and softmax classifier layers were added to the baseline EfficientNetV2-Small model. Fig. 4 shows the architecture of the proposed modified EfficientNet with softmax classifier. In the TL, the lower layers capture the general features essential for classifying DR and are retrained, whereas the higher layers are fine-tuned. This technique significantly

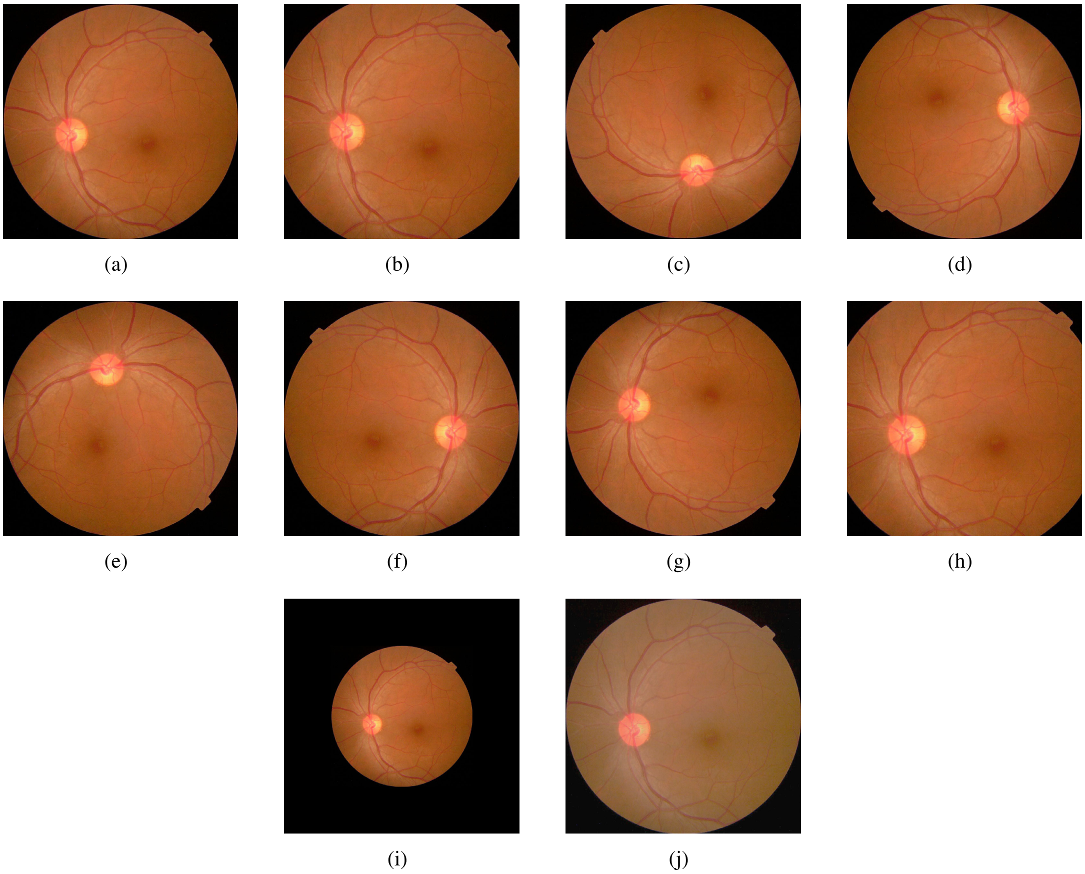

FIGURE 3. Different image augmentation techniques used during the training phase. (a) original image; (b) random rotation; (c) rotation of 90°; (d) rotation of 180°; (e) rotation of 270°; (f) horizontal flip; (g) vertical flip; (h) zoom-in; (i) zoom-out; (j) contrast adjustment.

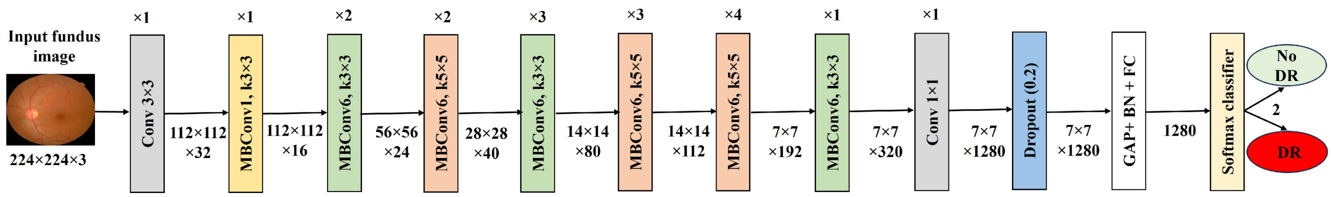

FIGURE 4. The architecture of the modified EfficientNet.

reduces computational resources and time compared to training the DL model from scratch. Table 1 lists the details of the additional layers integrated into EfficientNetV2-Small to build the modified EfficientNet model.

During the training phase of the modified EfficientNet, the first 10 layers from the top, excluding the addon layers of the EfficientNetV2-Small, were unfrozen, while the remaining layers were kept frozen. Because TL is used, a low learning rate is preferred to minimize changes to the network's pre-trained knowledge with ImageNet weights. Dropout and BN layers were placed before the FC layer to prevent overfitting. The model was then trained for 20 epochs. Early stopping [46] is a technique used to avoid overfitting. The training process was halted as soon as the validation loss increased in the following cycle. Table 2 shows the hyperparameter configurations used for training the modified EfficientNet.

D. SUPPORT VECTOR MACHINES (SVM) CLASSIFIER

In the second method, traditional feature extraction techniques such as Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), Color Histograms, and Local Binary Patterns (LBP) are used to convert input fundus images into feature vectors for training binary SVM classifier with an RBF kernel [47].

E. HYBRID MODEL

In the third method, a hybrid model called EffNet-SVM is designed using the modified EfficientNet and an SVM with an RBF kernel classifier. In this approach, the final fully

TABLE 2. Hyperparameter configuration used for training the modified EfficientNet.

Hyperparameters	Value
Image size	224 x 224
Dropout rate	0.2
Learning rate	0.0001
Batch size	32
Loss	Categorical cross entropy
Number of classes at the output layer	2
Optimizer	Adam
Number of epochs	20

connected (FC) layer of EfficientNet generates 1,280 feature vectors from input fundus images, capturing high-level patterns such as nerve edges, textures, and disease-specific structures. These features are then fed into the SVM classifier with RBF kernel, replacing the conventional softmax layer in baseline EfficientNet. The RBF kernel enables SVM with non-linear decision boundaries, enhancing classification accuracy for complex fundus images where classes are not linearly separable. This hybrid design leverages EfficientNet's efficiency in scaling and spatial feature extraction while benefiting from SVM's robustness in high-dimensional classification tasks [48].

Fig. 5 shows the architecture of the EffNet-SVM model. The EffNet-SVM model employs hyperparameters optimized for the EfficientNet, which utilizes TL with pre-trained ImageNet weights as shown in Table 2. At the same time, the SVM's regularization parameter C and kernel coefficient γ are tuned via grid search.

III. RESULTS AND DISCUSSION

This section presents a comparative analysis of the proposed EffNet-SVM model against eight state-of-the-art DL-based models for DR classification using fundus images, as reported in the literature. Additionally, the comparison includes the modified EfficientNet, the SVM classifier, and widely used CNNs such as MobileNetV2, ResNet50, and DenseNet-121. The performance evaluation is conducted using widely accepted metrics, including the confusion matrix, accuracy, precision, recall, the ROC curve, and AUC [49].

A. CONFUSION MATRIX AND ACCURACY

The confusion matrix is an essential tool for evaluating classification models, as it provides insights into the counts of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Specifically, a TP indicates instances where the model accurately predicts the presence of DR, representing a successful, positive prediction made by the classifier. An FP occurs when the model incorrectly predicts the presence of DR in cases where it is absent, thereby highlighting a false alarm in the prediction. A TN signifies instances where the model correctly predicts the absence of DR when it is absent, demonstrating a successful negative prediction. Conversely, an FN refers to instances where the model fails to predict the presence of DR when it is actually present, indicating a missed detection.

In the context of DR diagnosis, minimizing FN is particularly critical, as overlooking cases of DR can result in delayed treatment and irreversible vision loss; however, minimizing FP is also important. The confusion matrices for the modified EfficientNet, SVM (RBF), and the proposed EffNet-SVM, trained and tested on the APTOS dataset [10], are presented in Fig. 6.

From the confusion matrix, accuracy can be derived as the proportion of correct predictions (both TP and TN) out of the total number of samples in the data subset [50].

The accuracy formula is given as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

The confusion matrices for each model were analyzed to determine accuracy values. The computed accuracy for the three models is presented in Table 3. As shown in Table 3, the EffNet-SVM consistently outperforms the other two models, achieving the highest accuracy of 97.26%, followed by the Modified EfficientNet with an accuracy of 95.90%, while the SVM (RBF) demonstrates the lowest accuracy of 94.80%. These results indicate that EffNet-SVM is the most reliable model among the three in terms of overall classification performance.

B. PRECISION AND RECALL

Precision and recall are critical evaluation metrics, particularly when class imbalance is a concern. Precision measures the proportion of true positive predictions out of all positive predictions, while recall measures the proportion of actual positive instances correctly identified by the model. The precision and recall formulae are given in (2) and (3) respectively:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

The F1-score is the harmonic mean of precision and recall and is particularly useful when the class distribution is imbalanced. It can be computed using (4):

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

The accuracy, precision, recall, F1-Score, and precision-recall AUC for each model are shown in Table 3.

As detailed in Table 3, EffNet-SVM achieves superior precision (0.9698), recall (0.9796), and F1-score (0.9746), indicating an improved balance between minimizing FP and maximizing TP detection compared to both modified EfficientNet and SVM (RBF). The trade-off between precision and recall is particularly noticeable for SVM (RBF), which exhibits lower values for both metrics, potentially due to the limitations of the hand-engineered features used for training. This improved balance of EffNet-SVM minimizes the risk

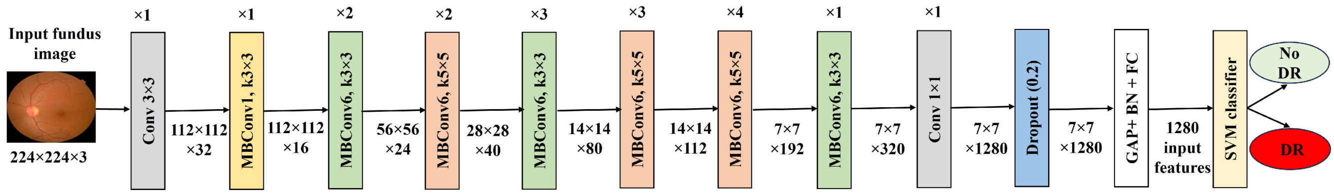

FIGURE 5. The architecture of the EffNet-SVM model.

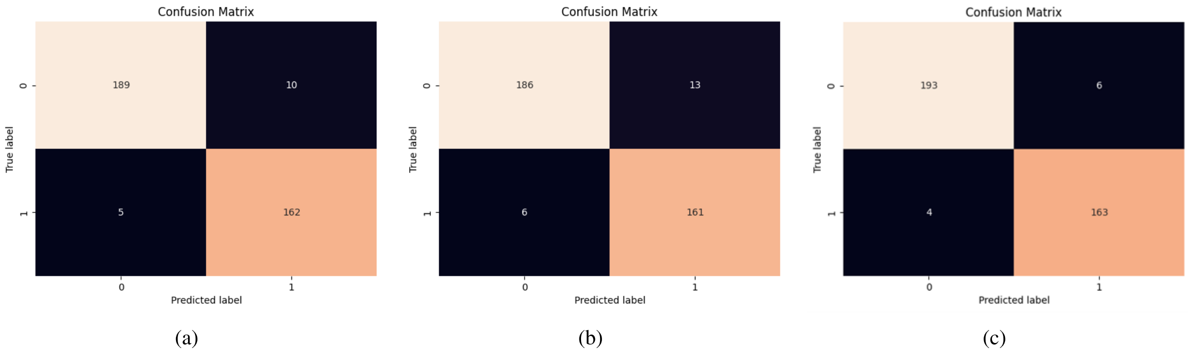

FIGURE 6. Confusion Matrices: (a) confusion matrix of modified EfficientNet; (b) confusion matrix of SVM classifier; (c) confusion matrix of EffNet-SVM classifier.

TABLE 3. Performance Comparison of EffNet-SVM against modified EfficientNet and SVM.

Model	Accuracy	Precision	Recall	F1-Score	Precision-Recall AUC
Modified EfficientNet	95.90	0.94	0.97	0.96	0.96
SVM (RBF)	94.80	0.93	0.96	0.95	0.95
EffNet-SVM	97.26	0.96	0.97	0.97	0.97

of both missed diagnoses and unnecessary referrals, leading to more effective and efficient DR screening. The use of feature extraction allows for improved diagnosis performance in clinical settings.

C. PRECISION-RECALL ANALYSIS AND AUC

Precision-recall analyses are particularly useful in imbalanced datasets, as they offer an in-depth analysis of how precision and recall interact across a spectrum of decision thresholds [51]. As shown in Table 3, EffNet-SVM again outperforms both Modified EfficientNet and SVM (RBF) in terms of precision-recall AUC, indicating its superior ability to maintain a high precision while detecting most positive instances. Modified EfficientNet shows good performance, though slightly lower than EffNet-SVM. At the same time, SVM (RBF) lags behind in both the precision-recall curve and the AUC, further underscoring its comparatively weaker classification ability.

Table 4 compares key performance metrics, network depth, the number of trainable parameters, and inference time of the proposed hybrid model against widely used CNN models for image classification. The comparison includes MobileNetV2 (a lightweight model), ResNet50 (a model with more parameters), and DenseNet-121 (a deeper network), trained and tested with the APTOS dataset [10].

As shown in Table 4, the proposed EffNet-SVM model outperforms benchmark CNN models, including MobileNetV2, ResNet50, and DenseNet-121, on the APTOS dataset [10]. The EffNet-SVM obtained the highest accuracy (97.26%) with high precision and recall (0.97% each) while maintaining a low inference time of 0.0982 seconds. This balance of performance, network depth (55 layers), and trainable parameters (20 million) underscores EffNet-SVM's suitability for high-performance applications and resource-constrained environments, making it an optimal choice for DR diagnosis.

D. GENERALIZATION OF THE MODEL PERFORMANCE

Table 5 presents the accuracy of three CNN models, MobileNetV2, ResNet50, and DenseNet121, against the proposed EffNet-SVM in a cross-dataset evaluation setting. The models were trained on the APTOS dataset [10] and tested on the HRF dataset [52] to assess their generalizability to unseen images. The HRF dataset [52] is valuable for its high-resolution images, comprising 15 images from healthy patients and 15 from patients with DR. In contrast, the APTOS dataset [10] contains no high-resolution images.

TABLE 4. Comparison of model performance metrics and computational efficiency on the APTOS dataset [10].

Model	Accuracy	Precision	Recall	F1-Score	Depth	Parameters (Millions)	Inference Time (s)
MobileNetV2	82.78	0.83	0.82	0.82	53	3.5	0.1023
ResNet50	91.25	0.92	0.91	0.91	50	23.5	0.1114
DenseNet-121	93.44	0.93	0.93	0.93	121	7.9	0.1197
EffNet-SVM	97.26	0.96	0.97	0.97	55	20	0.0982

TABLE 5. Cross-dataset evaluation: Accuracy of CNN models and EffNet-SVM trained on APTOS dataset [10] and tested on HRF image dataset [52].

Model	Accuracy
MobileNetV2	66.33
ResNet50V2	72.00
DenseNet-121	72.45
EffNet-SVM	76.76

Thus, this evaluation primarily measures the models' ability to generalize across datasets.

From Table 5, it can be observed that MobileNetV2 achieved the lowest accuracy (66.33%), suggesting potential challenges in adapting to a different dataset. ResNet50 and DenseNet-121 performed better, with accuracy scores of 72.00% and 72.45%, respectively. The proposed hybrid model, EffNet-SVM, outperformed all other models, achieving an accuracy of 76.76%, demonstrating superior generalization capabilities. These results underscore EffNet-SVM as a more reliable model for real-world applications involving unseen data.

E. GRADIENT-WEIGHTED CLASS ACTIVATION MAPPING (GRAD-CAM) ANALYSIS

Grad-CAM is used to visualize the regions of an image that a CNN model focuses on when making predictions. This analysis enhances the interpretability of the model's decision-making process. Grad-CAM generates a heatmap that highlights important areas of an image by leveraging the gradients of the target class from the final convolutional layer. An understanding of which specific areas of an input image a CNN attends to when generating its predictions is key to establishing trust. Grad-CAM analysis emphasizes closer correspondence to the targeted output variable as calculated by the final convolutional layer. Figs. 7, 8, 9, and 10 present the Grad-CAM analysis for MobileNetV2, ResNet50, DenseNet-121, and the proposed EffNet-SVM model.

From Fig. 7, it can be observed that on the test image-1 (a), the MobileNetV2 model correctly classified as no DR Image. Minimal activation is observed, consistent with the absence of DR features, resulting in an accurate prediction. On test image-2 (b), the MobileNetV2 model incorrectly classified as no DR. The heatmap shows misplaced activations, failing to capture key pathological areas, causing a misclassification. While on test image-3 (c), the MobileNetV2 model incorrectly classified as DR. The model activates around non-pathological regions, mistakenly identifying them as indicative of DR.

Fig. 8, it can be observed that on the test image-1 (a), ResNet50 correctly classified as no DR. The heatmap shows low activation, which is consistent with the absence of DR features. On test image-2 (b), ResNet50 incorrectly classified as no DR. The model overlooks key pathological areas, leading to an incorrect prediction. While on test image-3, the ResNet50 model could correctly be classified as DR as there is strong activation around abnormal regions.

Fig. 9, it can be observed that on the test image-1 (a), the DenseNet-121 model could correctly be classified as no DR with minimal activation in non-pathological regions. In test image-2, the model was correctly classified as having no DR. The model focused on relevant areas supports correct classification. On test image-3 (c), the model correctly classified as DR with strong activations around lesions, indicating effective identification of abnormalities.

From Fig. 10, it can be observed that on the test image-1, EffNet-SVM correctly classified as no DR. The Grad-CAM heatmap shows minimal activation around the optic disc and other regions, indicating that the model correctly identified the absence of significant pathological features associated with DR. In test image-2, EffNet-SVM incorrectly classified as having no DR. The heatmap highlights areas that do not correspond to critical pathological features of DR, suggesting the model focused on irrelevant parts, leading to a misclassification. On test image-3, EffNet-SVM correctly classified the image belonging to DR. The heatmap shows strong activation in areas with exudates, indicative of DR. This indicates the model successfully identified and focused on the key pathological features for accurate classification.

Table 6 presents a comparison of the proposed EffNet-SVM model with eight state-of-the-art DL-based methods for DR classification using fundus images, as reported in the literature. The comparison is based on accuracy, where EffNet-SVM demonstrates superior performance, achieving the highest accuracy of 97.26%, outperforming all eight DL models.

The comprehensive evaluation of the models consistently demonstrates that the hybrid EffNet-SVM model outperforms other DL and machine learning approaches, achieving the highest accuracy, AUC scores, and precision-recall performance. EffNet-SVM exhibits superior capability in distinguishing positive and negative instances while maintaining an optimal balance between precision and recall.

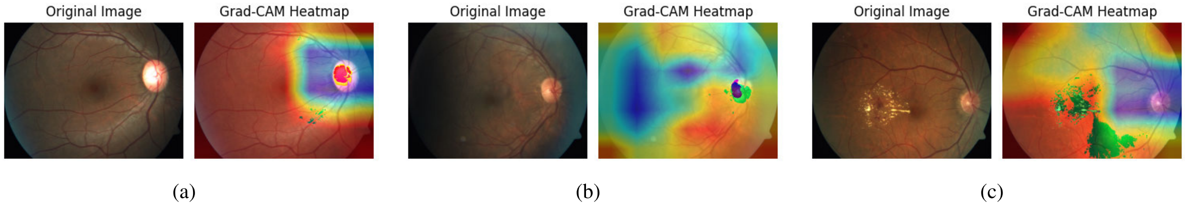

FIGURE 7. Grad-CAM analysis of MobileNetV2: (a) test image-1: correctly classified as no DR, (b) test image-2: incorrectly classified as no DR, (c) test image-3: incorrectly classified as DR.

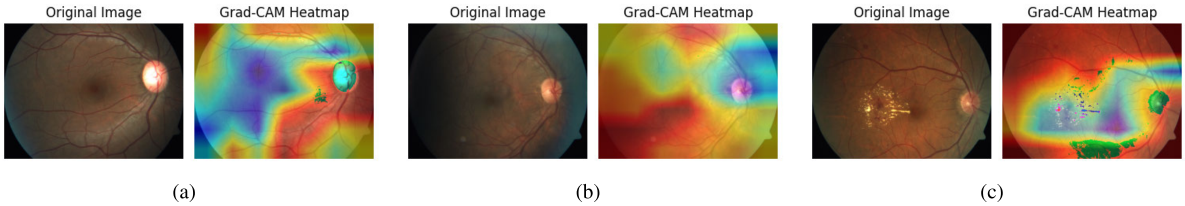

FIGURE 8. Grad-CAM analysis of ResNet50: (a) test image-1: correctly classified as no DR, (b) test image-2: incorrectly classified as no DR, (c) test image-3: correctly classified as DR.

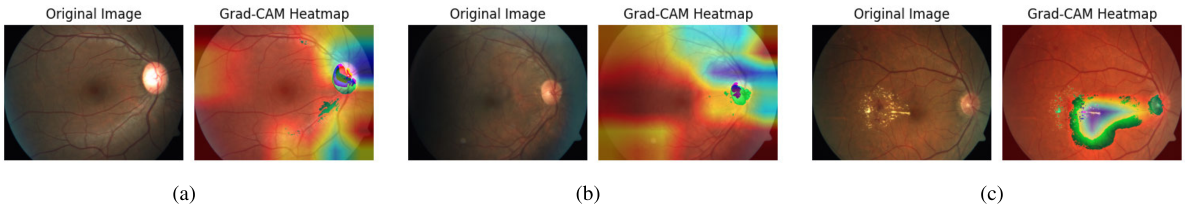

FIGURE 9. Grad-CAM analysis of DenseNet-121: (a) test image-1: correctly classified as no DR, (b) test image-2: correctly classified as DR, (c) test image-3: correctly classified as DR.

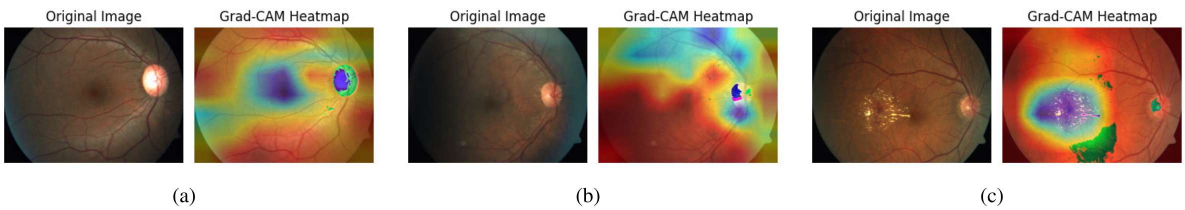

FIGURE 10. Grad-CAM analysis of EffNet-SVM: (a) test image-1: correctly classified as no DR, (b) test image-2: incorrectly classified no DR, (c) test image-3: correctly classified as DR.

TABLE 6. Comparison of the proposed EfficietNet-SVM against recent state-of-the-art DL models on the APTOS dataset [10].

Reference	Year	Method	Accuracy (%)
Taufiqurrahman et al. [31]	2020	MobileNetV2-SVM	85.00
Kobat at al. [33]	2022	Cubic SVM	85.93
Mandiga et al. [32]	2022	Fine tuned MobileNetV2-SVM	86.00
Padmanayana & Anoop [34]	2022	Custom CNN	94.60
Wiratama et al. [23]	2023	MobileNetV2	92.60
Mohanty et al. [24]	2023	VGG16 + XGBoost	79.50
Sajid at al. [53]	2023	NASNet	96.05
Taifa et al. [27]	2024	Ensemble models	95.50
		EffNet-SVM	**97.26**

These findings highlight the superiority of the EffNet-SVM hybrid model for DR classification and suggest its suitability for real-world applications where accuracy, precision, and recall are critical for early detection, enabling timely medical intervention.

IV. CONCLUSION

This paper introduces the EffNet-SVM hybrid model for classifying fundus images of both healthy eyes and those affected by DR. The model was trained and evaluated on the APTOS dataset, employing techniques such as TL, model fine-tuning, and image augmentation to enhance generalization and mitigate overfitting. EffNet-SVM's performance was compared against eight state-of-the-art DL models from the literature, demonstrating significant superiority overall. Additionally, the model exhibited strong data efficiency, maintaining high accuracy even with small datasets—a crucial advantage for DR screening in resource-limited rural settings. However, further validation on additional datasets is necessary to confirm its robustness.

While EffNet-SVM has shown strong feature learning capabilities, its performance is highly dependent on fundus image quality. Factors such as uneven illumination, low contrast, noise, and motion blur can compromise the accuracy of extracted features, potentially affecting classification performance. Although Grad-CAM analysis offers partial interpretability of features learned by EfficientNetV2-Small, the opaque decision-making process of the SVM classifier remains a significant limitation. This lack of end-to-end interpretability may hinder clinical adoption, as transparency in AI-driven decisions is crucial for clinician trust and actionable insights. Future research could explore interpretable attention mechanisms and image preprocessing pipelines to improve robustness against poor-quality fundus images. Additionally, a multimodal extension of EffNet-SVM could be developed to facilitate more precise DR severity grading by integrating complementary diagnostic information.

DATA AND CODE AVAILABILITY

Data and code are freely available at https://github.com/Boredlambda/EffNet-SVM

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